DEPARTMENT OF SOFTWARE ENGINEERING & RESEARCH

Final Year Capstone Project (CS499.12) Guidelines

1. Project Milestones and Timeline

The Capstone Project spans across two semesters (Year 4, Semester 1 & 2) and carries a weight of 12 academic credits. All students must strictly adhere to the submission deadlines outlined below:

Milestone	Target Deadline	Deliverable Required	Weightage
Proposal Defense	Week 4 (Sem 1)	Project Proposal & Literature Matrix	10%
Interim Review	Week 14 (Sem 1)	Intermediate Code Base & Progress Report	20%
Final Thesis	Week 10 (Sem 2)	Complete Hardbound Thesis (80-100 pages)	40%
Oral Defense	Week 12 (Sem 2)	Viva Voce Presentation & System Demo	20%
Supervisor Evaluation	Continuous	Weekly Logs & Professional Conduct	10%

2. Key Research Domains for 2026/2027

Students are strongly encouraged to select topics aligned with current departmental research focus areas:

- **Advanced Retrieval-Augmented Generation (RAG):** Hybrid search, query rewriting, reranking algorithms, vector database scalability.
- **Autonomous AI Agents:** Tool-calling frameworks, multi-agent collaboration protocols, and safety guardrails.
- **Edge AI & Quantization:** Efficient deployment of small language models (SLMs) on IoT hardware.

3. Minimum Requirements for Passing

To achieve a passing grade (Grade C or above), the submitted code must run without unhandled runtime crashes, achieve a minimum unit test coverage of 75%, and pass an automated originality scan with less than 15% similarity.